CocoConnect

Functional Analysis Document

Version 2 – Product Marketplace and Service Marketplace

Client: Coconut Growers Association of Sri Lanka · Group 38

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Document History

Version	Content
1	Requirement Analysis Document: stakeholders, current problems, consolidated and prioritised requirements, gaps between sources
2	This document: detailed analysis of the Product Marketplace and Service Marketplace – functions, frontend components, backend functions, database interactions, workflow consistency and confirmed requirements

1. Purpose

This document analyses the two marketplace modules of CocoConnect in detail. For each one it defines the functions, the screens and components the frontend needs, the backend functions and API endpoints, and the database tables involved. It then checks that the buyer and estate owner workflows match each other, and records which requirements were confirmed with stakeholders.

Requirement IDs refer to the Software Requirements Specification, Version 3.

2. Modules Analysed

Module	Actors	Goal	SRS v3
Product Marketplace	Buyer, Estate Owner / Seller	Buy and sell coconuts directly, agree a fair price, and track the order to completion	FR-08 to FR-16
Service Marketplace	Estate Owner, Estate Manager, Worker, Service Provider	Find and hire nearby workers and machinery for estate work	FR-17 to FR-23

3. Product Marketplace Analysis

3.1 Functions

ID	Function	Actor	SRS v3	Priority
PM-F1	Post a sell offer (product type, quantity, available date, price per nut)	Estate Owner	FR-08, FR-13	**Must**
PM-F2	Edit or remove own listings	Estate Owner	FR-09, FR-16	**Must**
PM-F3	Browse listings by coconut type (raw, husked, dehusked)	Buyer	FR-10, FR-14	**Must**
PM-F4	Filter and sort listings by price, distance, quantity and date	Buyer	FR-11	**Must**
PM-F5	View listing details (estate, location, quantity, date, price)	Buyer	FR-12	**Must**
PM-F6	Post a buy request (quantity, required date, bidding price)	Buyer	FR-11, FR-15	**Must**
PM-F7	View matching buyer requests and send an offer	Estate Owner	FR-15	**Must**
PM-F8	Negotiate a price in chat; accept or reject offers	Buyer, Estate Owner	FR-15, FR-40	**Must**
PM-F9	Create an order when an offer is accepted	System	FR-15	**Must**
PM-F10	Complete a deal with the delivered quantity and final price	Estate Owner	BR-02	**Must**
PM-F11	Show market price information (latest deal, average price)	Estate Owner	–	**Should**
PM-F12	Rate the other party after a completed order	Buyer, Estate Owner	FR-34, FR-35	**Should**

3.2 Frontend components

Screen / component	Used by	Purpose
Buying Categories	Buyer	Entry point: choose Coconuts, Vegetables, Livestock or Other Goods

Screen / component	Used by	Purpose
Coconut Categories	Buyer	Choose raw, husked or dehusked coconuts
Marketplace (request form + listing list)	Buyer	Post a buy request and browse nearby listings
Listing card	Buyer	Estate, distance, price, quantity, date, trust indicator
Filter sheet / distance chips	Buyer, Estate Owner	Within 10, 25 or 50 km; sort by price or date
Listing detail	Buyer	Full listing information and Negotiate button
Estate Owner Dashboard	Estate Owner	Entry point: selling categories and provider services
Sell screen (Sell Summary + Interested Buyers)	Estate Owner	Post a sell offer and see matching buyer requests
Buyer request card	Estate Owner	Buyer, quantity, needed-by date, bidding price, Send Offer
Negotiation chat with offer cards	Both	Messages and price offers with Accept / Reject
Orders (Pending / Deals / Transactions)	Both	Track requests, active deals and completed orders
Complete Deal form	Estate Owner	Enter delivered quantity and final price
Market price strip	Estate Owner	Latest closed deal and average price per nut
Bottom navigation	Both	Dashboard, Orders, Chats, Alerts, Profile

3.3 Backend functions and API endpoints

Endpoint	Function	Role
`GET /api/buyer/listings`	List listings; filter by type, quantity, date, distance; sort	Public
`GET /api/buyer/listings/:id`	One listing's details	Public
`POST /api/buyer/requests`	Create a buy request	Buyer
`GET, PUT, DELETE /api/buyer/requests/:id`	View, edit or delete own requests	Buyer
`POST /api/owner/listings`	Post a sell offer	Estate Owner
`GET /api/owner/buyer-requests`	Open buyer requests matching the estate's harvest	Estate Owner
`POST /api/owner/sell-requests`	Send an offer on a buyer request	Estate Owner
`GET /api/chats, GET /api/chats/:id`	List chats; open one	Both
`POST /api/chats/:id/messages`	Send a message or price offer	Both
`POST /api/chats/:id/offers/:msgId/accept | reject`	Answer the other side's offer	Both
`POST /api/owner/orders/:id/complete`	Complete a deal	Estate Owner
`GET /api/owner/market-stats`	Latest deal and average price	Estate Owner

3.4 Database interactions

Action	Tables read	Tables written
Browse listings	product_listings, estates, users (location)	–
Post a buy request	–	buyer_requests
Post a sell offer	estates	product_listings
Send an offer on a request	buyer_requests	sell_requests, negotiations, negotiation_messages
Send a chat message or offer	negotiations	negotiation_messages
Accept an offer (one transaction)	negotiations, negotiation_messages	orders, negotiations, negotiation_messages, sell_requests, buyer_requests
Complete a deal	orders	orders (delivered quantity, final price, status)
Market price	orders, product_listings, buyer_requests	–

3.5 Rules and validation

- Quantity and price must be greater than zero; the required date cannot be in the past.
- A listing matches a request when it has at least the requested quantity and is available on the required date or up to 7 days after it.
- An estate can send only one offer per buyer request.
- Only the receiving side can accept or reject an offer; a newer offer replaces the pending one.
- Accepting an offer creates the order and closes the other estates' offers on the same request in one database transaction.
- A final price that differs from the agreed price must be reviewed by an administrator.
- Exact coordinates are never sent to other users; only the distance is shown.

4. Service Marketplace Analysis

4.1 Functions

ID	Function	Actor	SRS v3	Priority
SM-F1	Create a service profile (skill type, charging method per tree or per day, rate, availability)	Worker	FR-17, FR-20	**Must**
SM-F2	List machinery with a daily rate and availability	Service Provider	FR-19	**Must**
SM-F3	Search workers and machinery by category	Estate Owner, Estate Manager	FR-18	**Must**
SM-F4	Find providers near the estate on a map, with rates	Estate Owner	FR-18	**Should**
SM-F5	Send a hiring request with date and estate	Estate Owner	FR-21	**Must**
SM-F6	Accept or reject a hiring request	Worker, Service Provider	FR-22	**Must**
SM-F7	Track job status (requested, accepted, in progress, completed)	All	FR-23	**Must**
SM-F8	View booked services on a calendar	Estate Owner	–	**Should**
SM-F9	Notify both sides of new, accepted and rejected requests	System	FR-38, FR-39	**Must**
SM-F10	Rate the worker and the estate owner after a completed job	Estate Owner, Worker	FR-32, FR-33	**Should**

4.2 Frontend components

Screen / component	Used by	Purpose
Provider Services section on the dashboard	Estate Owner	Entry to Fertilizer, Machinery, Labour and other services
Service category list	Estate Owner	Plucking, husking, fertilizing, gathering, loading, tractor services, equipment
Machinery map	Estate Owner	Nearby tractors and machines as map markers with daily rates
Provider card	Estate Owner	Name, skill, charging method, rate, distance, rating, Request button
Hiring request form	Estate Owner	Service, date, estate, number of trees or days, note
Service schedule calendar	Estate Owner	Month view with confirmed and pending bookings
Provider dashboard and job requests list	Worker, Service Provider	Incoming requests with Accept / Reject
Service profile form	Worker, Service Provider	Create and edit the advertised service

4.3 Proposed backend functions and API endpoints

Endpoint (proposed)	Function	Role
`POST /api/services/profiles`	Create or update a service profile	Worker, Provider
`POST /api/services/equipment`	List a machine with daily rate	Provider
`GET /api/services/providers?category=&radiusKm=`	Search providers by category and distance	Estate Owner
`POST /api/services/requests`	Send a hiring request	Estate Owner
`POST /api/services/requests/:id/accept | reject`	Answer a hiring request	Worker, Provider
`GET /api/services/requests`	My requests and their status	All
`GET /api/services/schedule?month=`	Bookings for the calendar	Estate Owner

4.4 Database interactions

Action	Entities (ER diagram)
Create a service profile	Service_Profile linked to Worker
List machinery	Equipment linked to Service_Provider
Search providers	Service_Profile, Equipment, User (location)
Send a hiring request	Hiring_Request linked to Estate_Owner
Accept a request	Hiring_Request status updated; Job_Assignment created
Complete a job	Job_Assignment (end date, outcome, total pay); Rating
Calendar	Job_Assignment start and end dates

4.5 Rules and validation

- Workers must choose a service category and a charging method (per tree or per day) (BR-04).
- A job exists only after the worker accepts the hiring request (BR-05, BR-06).
- A provider cannot be booked twice for the same time.
- Ratings can be given only after a completed job (BR-07).
- The service marketplace reuses the product marketplace's patterns: distance filtering, request and accept flow, notifications and the shared chat.

5. Buyer and Estate Owner Workflow Consistency

Every action on one side must have a matching view or action on the other. The two workflows were compared step by step.

Step	Buyer side	Estate owner side	Result
1	Posts a buy request	Sees it under Interested Buyers if it matches the harvest	**Consistent**
2	Sees nearby listings	Posts a sell offer that appears in the marketplace	**Consistent**
3	Matching rule: listing available on the required date or up to 7 days after	Matching rule: request needed between 7 days before and the available date	**Consistent**
4	Taps Negotiate on a listing	The chat appears in Chats with an alert	**Consistent**
5	Receives the estate's offer on a request	Sends an offer, which opens a chat	**Consistent**
6	Accepts or rejects an offer	Accepts or rejects the buyer's counter-offer	**Consistent**
7	Sees the order in Orders → Deals with the due date	Sees the same deal with the same due date	**Consistent**
8	No action to complete	Completes the deal with delivered quantity and price	**Consistent**

Step	Buyer side	Estate owner side	Result
9	Sees the completed order in Transactions	Sees it in Transactions	**Consistent**
10	Rates the estate owner	Rates the buyer	**Gap found**

Gap: ratings are required on both sides (FR-34, FR-35) but neither workflow has a rating step yet. Add a rating prompt after a deal is completed.

6. Requirements Confirmed in Technical Review

Topic	Decision	Status
Coconut types	Raw, husked and dehusked, chosen before the marketplace	**Confirmed**
Selling unit	Price per nut	**Confirmed**
Price agreement	Negotiated in an in-app chat with offer cards, not by phone	**Confirmed**
Location	Location picked at sign-up; marketplace filtered by distance	**Confirmed**
Request matching	Quantity at least the request; date within 7 days	**Confirmed**
Deal completion	Estate owner records delivered quantity and final price; price changes reviewed by admin	**Confirmed with change**
Navigation	Same bottom bar for both roles: Dashboard, Orders, Chats, Alerts, Profile	**Confirmed**
Service marketplace	Map of nearby machinery with rates, plus a service calendar	**Confirmed**
Online payment	Payment stays outside the app in this release	**Deferred**

7. Implementation Feasibility Summary

Area	Assessment
Product Marketplace	High: standard database operations and REST endpoints; transactions keep deals consistent
Negotiation chat	High: messages stored in the database; the app refreshes to show new messages
Location filtering	High: coordinates stored per user; distance calculated on the server
Service Marketplace	High: reuses the marketplace patterns; map uses free OpenStreetMap tiles
Service calendar	Medium: needs booking dates and checks for double bookings
Ratings	High: simple records linked to completed orders and jobs